Question:1 Create a DataFrame from the following dictionary: python CopyEdit data = { 'Name': ['Alice', 'Bob', 'Charlie', 'David'], 'Age': [25, 30, 35, 40], 'City': ['New York', 'Los Angeles', 'Chicago', 'Houston'] } Display the DataFrame.

```
import pandas as pd
data = {
'Name': ['Alice', 'Bob', 'Charlie', 'David'],
'Age': [25, 30, 35, 40],
'City': ['New York', 'Los Angeles', 'Chicago', 'Houston']
}
print(data);
```

```
{'Name': ['Alice', 'Bob', 'Charlie', 'David'], 'Age': [25, 30, 35, 40], 'City': ['New York', '
```

```
df=pd.DataFrame(data)
print(df);
```

```
      Name  Age         City
0    Alice   25     New York
1      Bob   30  Los Angeles
2  Charlie   35      Chicago
3    David   40      Houston
```

Question:2 From the above DataFrame, filter out the rows where the age is greater than 30.

```
filter=df[df['Age']>30]
print(filter)
```

```
      Name  Age     City
2  Charlie   35  Chicago
3    David   40  Houston
```

Question:3 Add a new column Salary with values [50000, 60000, 70000, 80000] to the existing DataFrame.

```
df['Salary']=[50000,60000,70000,80000]
print(df)
```

```
      Name  Age         City  Salary
0    Alice   25     New York   50000
1      Bob   30  Los Angeles   60000
2  Charlie   35      Chicago   70000
3    David   40      Houston   80000
```

Question:4 Increase the salary of all people by 10%.

```
df['Salary']=df['Salary']*1.10
print(df)
```

```
      Name  Age         City   Salary
0    Alice   25     New York  55000.0
```

```
1      Bob   30  Los Angeles  66000.0
2  Charlie   35      Chicago  77000.0
3    David   40      Houston  88000.0
```

Question:5 Remove the City column from the DataFrame.

```
df=df.drop('City',axis=1)
print(df)
```

```
      Name  Age   Salary
0    Alice   25  55000.0
1      Bob   30  66000.0
2  Charlie   35  77000.0
3    David   40  88000.0
```

Question:6 Rename the column Age to Years.*italicized text*

```
df=df.rename(columns={'Age':'Years'})
print(df)
```

```
      Name  Years   Salary
0    Alice     25  55000.0
1      Bob     30  66000.0
2  Charlie     35  77000.0
3    David     40  88000.0
```

Question:7 Sort the DataFrame by Salary in descending order.

```
df = df.sort_values(by='Salary', ascending=False)
print(df)
```

```
      Name  Years   Salary
3    David     40  88000.0
2  Charlie     35  77000.0
1      Bob     30  66000.0
0    Alice     25  55000.0
```

data = { 'Department': ['HR', 'IT', 'HR', 'IT'], 'Employee': ['Alice', 'Bob', 'Charlie', 'David'], 'Salary': [50000, 60000, 55000, 65000] } Group by department and find the average salary.

```
import pandas as pd

data = {
    'Department': ['HR', 'IT', 'HR', 'IT'],
    'Employee': ['Alice', 'Bob', 'Charlie', 'David'],
    'Salary': [50000, 60000, 55000, 65000]
}

df = pd.DataFrame(data)

average_salary = df.groupby('Department')['Salary'].mean()

print(average_salary)
```

```
Department
HR    52500.0
IT    62500.0
Name: Salary, dtype: float64
```

Question: Given two DataFrames: python CopyEdit df1 = pd.DataFrame({ 'ID': [1, 2, 3], 'Name': ['Alice', 'Bob', 'Charlie'] }) df2 = pd.DataFrame({ 'ID': [1, 2, 4],

'Score': [85, 90, 88] }) Merge them on the ID column.

```
import pandas as pd
df1 = pd.DataFrame({
'ID': [1, 2, 3],
'Name': ['Alice', 'Bob', 'Charlie']
})
df2 = pd.DataFrame({
'ID': [1, 2, 4],

'Score': [85, 90, 88]
})
merged_df = pd.merge(df1, df2, on='ID')
print(merged_df)
```

```
   ID   Name  Score
0   1  Alice     85
1   2    Bob     90
```